

JeongHyeon Kim

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RESEARCH INTEREST

My research interests lie in **humanoid robotics** and **whole-body control**. I am interested in developing controllers that enable humanoid robots to perform complex, dynamic tasks through the integration of model-based and learning-based approaches.

EDUCATION

- Korea University (GPA: 3.78/4.0, 4.25/4.5)** Mar.2025 – Present
M.S. in Mechanical Engineering Seoul, Republic of Korea
- Korea University (GPA: 3.48/4.0, 3.9/4.5)** 2019 – 2025
B.S. in Mechanical Engineering Seoul, Republic of Korea

EXPERIENCE

- Human-Machine Systems Lab, Korea University** Mar.2025 – Present
Graduate Research Student (M.S.) Seoul, Republic of Korea
– Research on humanoid robotics and whole-body control
- Human-Machine Systems Lab, Korea University** Mar.2024 – Feb.2025
Undergraduate Research Student Seoul, Republic of Korea
– Participated in robotics research projects

PROJECTS

- Quadruped Robot Obstacle Avoidance (CCP) — Team Leader** Mar.2024 – Jan.2025
Computer Vision, Autonomous Navigation
– Developed obstacle detection and avoidance pipeline for DOGZILLA quadruped robot
– Implemented monocular ORB-SLAM for depth estimation and UV-disparity for obstacle segmentation
- Ladder-Crossing Robot — Team Leader** Mar.2023 – Sep.2023
Mechanism Design and Control
– Designed the ladder-crossing mechanism and implemented the Arduino-based control code
- Solar Tracking Parasol — Team Leader** Sep.2023 – Nov.2023
Mechanism Design and Control
– Designed the link mechanism and implemented the sun-tracking control code

TECHNICAL SKILLS

Languages: Python, C/C++, MATLAB
Robotics: ROS2, Gazebo, URDF/XACRO
Tools: OpenCV, Arduino, Creo, NX, AutoCAD, SolidWorks, Git
OS: Linux (Ubuntu)

HONORS AND AWARDS

Fellowships and Scholarships

- Master's Research Encouragement Grant**, National Research Foundation of Korea 2025
Awarded to selected research-track graduate students for research funding support

Awards

- President's Award**, Creative Challenger Program (CCP), Korea University Jan.2025
Awarded for outstanding performance in autonomous obstacle avoidance for a quadruped robot
- Bronze Award**, Creative Comprehensive Design Competition, Korea University Sep.2023
Awarded for the Solar Tracking Parasol mechanism design project
- Design Award**, Capstone Design Fair, Korea University Nov.2023